

Mohammad hadi Niknam

Computer Science

Amirkabir University of Technology, Iran

+98-9918017161

niknam@aut.ac.ir

Twitter/X

Github Pages

GitHub Profile

HuggingFace

LinkedIn Profile

YouTube Channel

EDUCATION

- **Amirkabir University of Technology (Tehran Polytechnic)** 2023 – 2027 Feb.
Bachelor of Science in Computer Science 220 ECTS - 7th Semester – CGPA: 16.60/20
 - * **BSc Thesis:** *Human-Adaptive Safe Reinforcement Learning for Medical Ultrasound Robotic Arms*
 - * **Supervisor:** Prof. Mostafa Shamsi, Dr. Farid Azampour
- **Alborz High School** 2016 – 2022
High School Diploma in Mathematics & Physics (Previously American School) CGPA: 19.56/20

RESEARCH INTERESTS

- **Reinforcement Learning :** RL Theory, LLM Alignment, Decision Making Under Uncertainty, MultiAgent RL
- **Medical Robotics:** Robot learning, human-adaptive control, ultrasound-guided robotics
- **Medical AI:** Medical imaging, foundation models, clinically robust AI

KEY PROJECTS & RESEARCH

- * **Dr.Javad Alirezaie Lab** Toronto Metropolitan University - Canada - Remote
Research Assistant July 2025 - September 2026
 - URLSR : Efficient Image Super Resolution Network using Neural Operations and Reinforcement Learning**
(Under Review) (BMC Medical Informatics and Decision Making)
 - SRIQL: Implicit Q Learning for Ultralight-weight Image Super Resolution**
International Conference on Acoustics, Speech, and Signal Processing (ICASSP) | IEEE Conference
 - *Designed RL-based neural architecture search over lightweight SR blocks
 - ***What Was the Result:** SOTA on DIV2K / Set14 / Set5 / Urban100 by 1db PSNR
 - ***What I learn:** Each RL algorithm and environment adapt different based on Bias, Variance or TD error
- * *Research Project Coordinator* September 2025 - Now
 - Led a 3-member undergraduate research team on efficient image super-resolution, coordinating paper discussions, experimental milestones, task allocation, and research deliverables.
 - QuantaSR: Efficient Image Super-Resolution through Quantization Error Optimization in Low-Precision Neural Networks** [UnderReview]
 - **Efficient Super Resolution Using Multi-Task Learning with Conflict Aware optimization (MLTSR)** preparation expected by September-October
- * **Shafa Yahyaian Hospital, Iran University of Medical Sciences** Tehran, Iran
Research Assistant May 2025 – Present
 - **Publications / Manuscripts:**
 - Imaging-Based Artificial Intelligence for Osteonecrosis of the Jaw: A Systematic Review of Current Evidence and Clinical Translation Challenges — *Under Review*.
 - Fairness-Aware Operating-Room Sequencing for Acute Orthopaedic Trauma: A Retrospective Capacity-Calibrated Stress Test Using MOVER — *Under Review*, *BMC Medical Informatics and Decision Making*.
 - **Ongoing Research Proposals:**
 - BoneRADS framework automation with AI using classical computer vision
 - Fine-tuning MedSAM on the BTXRD dataset
 - VQA benchmark development for the BTXRD dataset
- * **Dr.Nassir Navab** Computer Aided Medical Procedures / TUM / Munich - Germany
Research Assistant May 2025 - Now
 - A Review on Ultrasound foundation model
- * **Dr. Gholamali Aminian** Remote
Research assistant September 2025 – August 2026
 - **LMArena140K: Human Preference Analysis** Investigated feature-level patterns associated with human preferences in large language model evaluations.

INDEPENDENT RESEARCH WORKS

- * **Independent Research** 2026 – Present
Reinforcement Learning
 - TailRL Extension (RLC 2026):** Designing an adaptive tail-weighting scheme to dynamically scale policy gradients based on trajectory return percentiles rather than heuristic top-*k* cutoffs.

WORK EXPERIENCE

- **ResanAI** *Tehran*
AI Research Engineer March 2026 – June 2026
- Completed an AI engineering internship focused on graph algorithms and Graph Neural Networks (GCN, GraphSAGE, and EdgeConv, a graph-based architecture and geometric algorithms in point cloud literature, Study PyG library internals

HONORS AWARDS

- * Gold medalist in Iran national Economics Olympiad
- * **Iran University Entrance Exam** : top 1% over 100,000 Iranian students compete for top universities.

SELECTED COURSES

- * PDE : (18.5/20), Signal&System : (18/20), Deep RL 2025 (audited), MultiAgent RL 2026 (audited)

LANGUAGES



Persian: Native Speaker



English: Professional Proficiency — Official TOEFL iBT: 90 / 120 (Verified Score in Hand)



German: Limited working Proficiency (B1) "*Goethe Institute Teheran*"



French: Limited understanding (A2)



Arabic: Basic understanding

PROFESSIONAL REFERENCES

- * **Dr. Farzan Niknejad** *fniknejad@torontomu.ca*
Toronto Metropolitan University - Canada Postdoctoral Fellow
- * **Dr. Alireza Esmaeilzahi** *alireza.esmaeilzahi@utoronto.ca*
University of Toronto - Canada Postdoctoral Fellow / PhD
- * **Dr. Mohammad Farid Azampour** *mf.azampour@tum.de*
Technical University of Munich - Germany Postdoctoral Researcher
- * **Dr. Mohammad Reza Dolikhani** *mohammadrezadolikhani@sbmu.ac.ir*
Shahid Beheshti University of Medical Sciences - Iran Medical Doctor (MD) - Researcher
- * **Dr. Javad Alirezaie** *javad@torontomu.ca*
Toronto Metropolitan University - Canada Professor

Latest version available at:
<https://www.overleaf.com/read/hdrpxqfmkwfg#14702d>